

Unit-1

1. What is a digital image? Draw and explain the block diagram of a typical image processing system in brief. (1+5) [2075]
2. How many images of size 1024×768 with 256 gray levels can be stored in a 2048 MB storage space? (5) [Model Q]
3. Discuss the various applications and problems associated with digital image processing in brief. (5) [2076]

Unit-2

1. Explain "power law transformation" techniques for the purpose of image enhancement. Explain the mean filter along with suitable algorithm for its implementation. (4+6)
2. Discuss the algorithm for histogram equalization.
3. Explain the first derivative filter with a suitable example.
4. Discuss the magnification of image using interpolation technique.
5. What is an edge detection filter? Differentiate between the first derivative and second derivative filter? Derive the filter mask for Laplacian filter and write the algorithm for its implementation.
[1+2+7=10]
6. Explain the term Contrast Stretching and the Histogram Equalization.
(3+1)
7. Derive the equation for Laplacian filter and write the algorithm for its implementation. (6)
8. Explain the Bit plane slicing technique for image enhancement.

Unit-3

1. Differentiate between Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT). Explain the FFT algorithm for one-dimensional case.

[3+7=10]

2. Describe in brief how do you implement Gaussian High Pass Frequency domain filter for image smoothing in the frequency domain?

3. What is Fourier Transform and how can you apply it in digital image processing? Explain the different properties of the Fourier Transform.

(4+6)

4. Define Discrete Cosine Transform (DCT). Differentiate between Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT).

5. How will you implement Butterworth high Pass Frequency domain filter for image sharpening in the frequency domain? Describe in brief.

Unit-4

1. Construct Huffman code for each gray level. Calculate the compression ratio and the relative data redundancy assuming if 3-bit code is used to code the gray level instead of Huffman code.

$$[7+3=10]$$

2. Explain lossy predictive coding in brief.
3. Describe lossless predictive coding model with a suitable block diagram.
4. Explain Contra-harmonic Mean Filters used for image restoration.
5. Explain the contra-harmonic mean filters used in image restoration.
6. What do you mean by Lossy Predictive Coding? Explain it with a suitable block diagram. (6)
7. Construct Huffman code for each gray level given and find the compression ratio and coding efficiency.

Gray Level	0	1	2	4	5	6	7
No. of Pixels	30	35	38	15	10	38	80

Unit-5

1. Write short notes on dilation and erosion.
2. Explain opening and closing morphological operations in brief.
3. Define region and Boundary. List some elements of Visual Perception.

[2+3]

Unit-6

1. Explain how Hough transform is useful in line detection?
2. Explain the region growing technique for image segmentation.
3. Explain the adaptive thresholding and region split and merge techniques for image segmentation. (5+5)
4. Explain in detail the region growing techniques for image segmentation. List the problems associated with region growing technique. (4+2)
5. Explain in detail the region split and merge techniques for image segmentation. List the problems associated with region split and merge technique. (4+2)
6. What is a gradient filter? Explain the Sobel gradient filter in detail along with its algorithm for implementation. (1 + 5)
7. Explain the segmentation by thresholding. Explain how you can apply segmentation in line detection.
8. How edges are detected using the gradient operators? Explain with suitable example.

Unit-7

1. Explain how a neural network can be applied in digital image processing with the help of a simple perceptron.
2. Discuss Neural Network based image recognition system with the help of a simple perception.
3. What are components of a pattern recognition system? Explain the relevant diagrams.
4. Write short notes on: (3 + 3)
 - a) Pattern Recognition and its applications
 - b) Neural Network
5. What is pattern and pattern recognition? Explain the steps and application areas of pattern recognition system.
6. What is pattern recognition? Explain the applications of neural networks in pattern recognition.